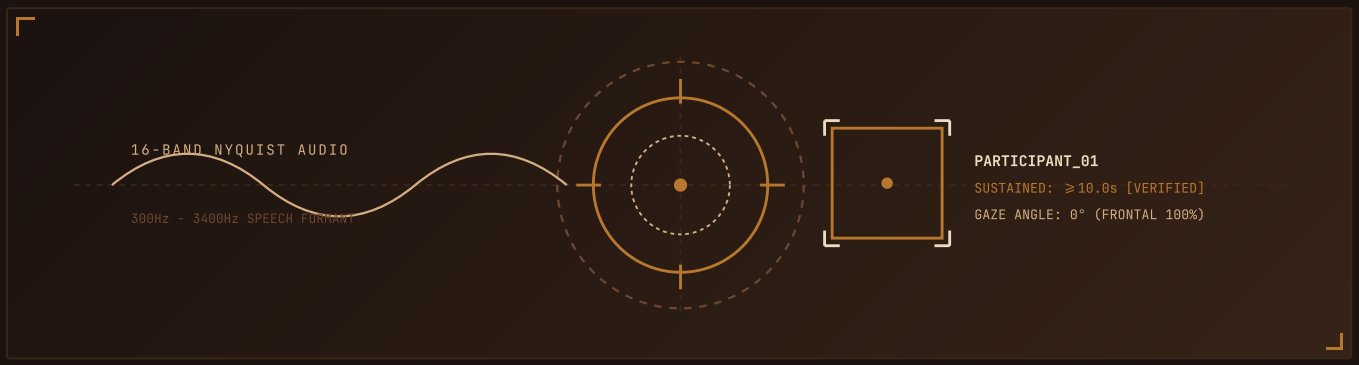
OPERATOR FIELD MANUAL & TUTORIAL COMPENDIUM

ATELIER // PRESENCE

Multimodal Cognitive Telemetry, Euclidean Centroid Tracking & Pedagogical Pacing Guide

EDITION: SLOW MORNING COFFEE EDITORIAL | CORE: OPENCV 4.9 + LIBROSA | FASTAPI + SQLITE WAL | AESTHETIC V2.1



00 / PHILOSOPHY

The Studio Seminars Philosophy

Traditional attendance software operates like cold corporate surveillance. **Atelier // Presence** is conceived as an architectural drawing companion: quiet, respectful, and non-invasive.

It calculates mathematical presence vectors, head-pose attention, and acoustic resonance purely in volatile memory. *Zero student facial photos or biometric scans are ever stored or uploaded.*

SYSTEM PREREQUISITES

Host Rig Specifications

Operating System	Windows 10/11, macOS, Linux
Python Environment	Python 3.11+ (.venv isolated)
Browser Client	Chromium / Chrome / Edge 110+
Sensory Hardware	USB/Built-in Webcam & Microphone
Data Storage Mode	SQLite WAL Mode (Localhost Only)

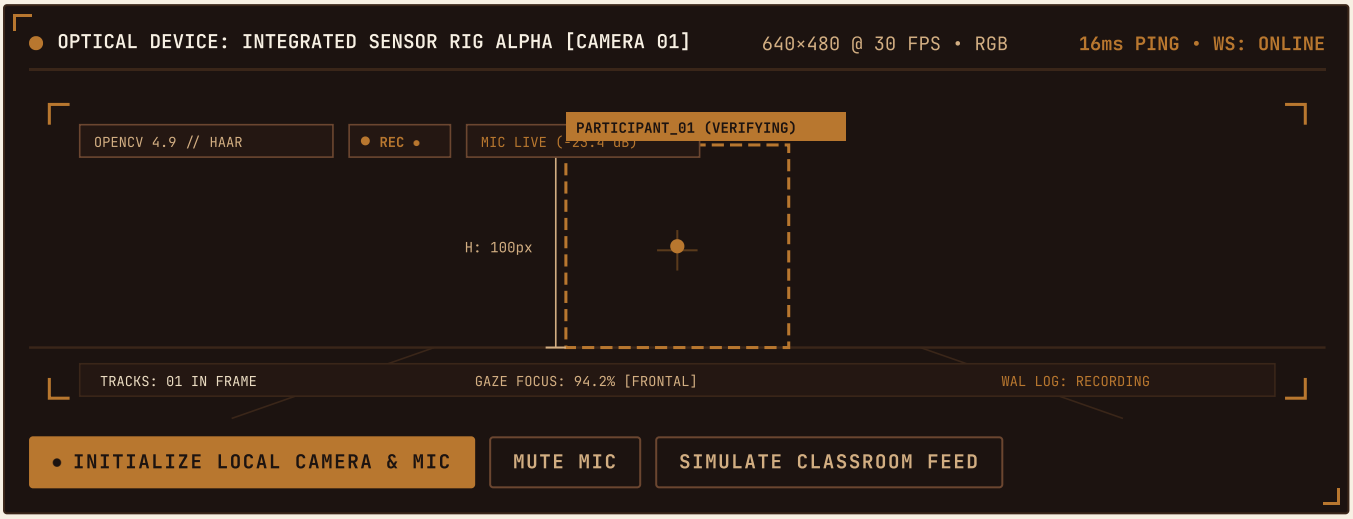
OPERATOR ITINERARY // CURATED AGENDA

SEC 01 Hardware Camera & Mic Initialization ..... Page 02
SEC 02 Euclidean Tracking & The 10s Rule ..... Page 03
SEC 03 16-Band Analog Spectrum Visualizer ..... Page 04
SEC 04 Classroom Simulation (No Webcam) ..... Page 05
SEC 05 Engine Calibration & CSV Export ..... Page 06
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STEP-BY-STEP OPERATIONAL MILESTONE 01

Initializing Hardware Camera & Microphone

Establish optical video acquisition and high-resolution 16 kHz WebAudio PCM streams with zero external browser plugins.



1 Trigger Hardware Initialization

In Section 03 (*The Live Telemetry Desk*), click the prominent caramel action button: `[Initialize Local Camera & Mic]`. This dispatches an asynchronous WebRTC acquisition call to the browser engine.

2 Grant Browser Media Permissions

When prompted by Chromium (*"localhost wants to Use your camera & microphone"*), select **Allow**. Processing occurs entirely in volatile memory; zero raw video or audio files are written to disk.

3 Verify Hardware Telemetry Strip

Inspect the status header directly above the viewfinder canvas:

- **Device Label:** Confirms active camera hardware name.
- **Resolution:** Locks to `640×480 @ 30 FPS RGB` for zero-latency inference.
- **WebSocket Ping:** Displays round-trip latency (<25ms).

4 Microphone Privacy Control

Toggle the tactile `[Mute Mic]` button to silence audio input whenever private conversations take place. The status updates instantly to `[MIC MUTED]`, pausing RMS and 16-band Nyquist calculations.

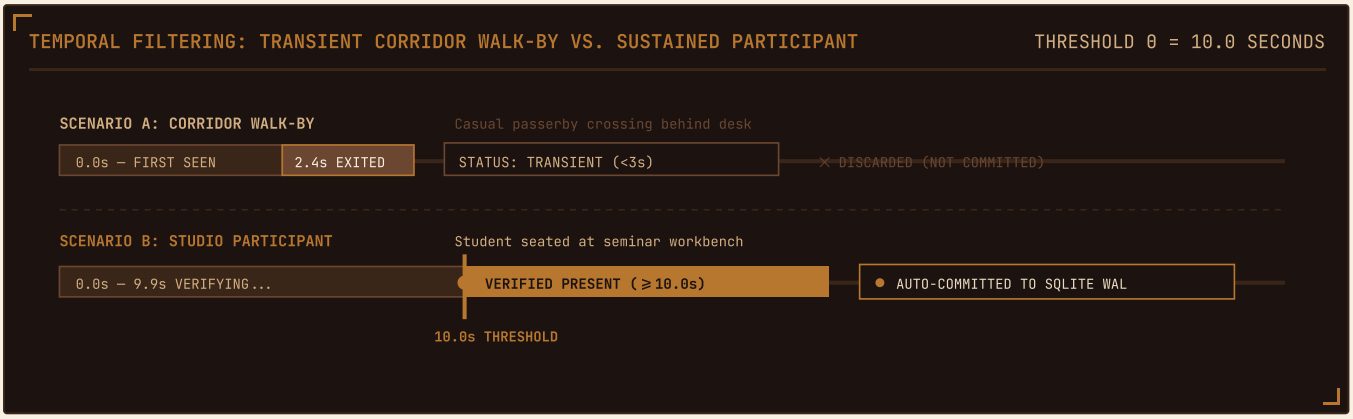
TECHNICAL NOTE // WEBAUDIO & OPENCV FRAME PIPELINE

Video frames are downsampled client-side to 640×480 and transmitted over binary WebSockets (`/ws/stream`) alongside 32-bit float PCM audio buffers. The backend processes frames with OpenCV Haar Cascades and Librosa spectral analysis, returning structured JSON metrics in <18ms.

STEP-BY-STEP OPERATIONAL MILESTONE 02

Euclidean Tracking & The 10-Second Attendance Rule

Understand mathematical centroid gating, transient rejection filters, and automated SQLite attendance commits.



MATHEMATICAL GATING

Euclidean Centroid Association

When a face bounding box (x, y, w, h) is detected, the engine calculates its central focal coordinate:

```
cX = x + (w / 2),    cY = y + (h / 2)
```

The tracker calculates the Euclidean distance between existing tracked participant centroids and new candidate detections. If distance $D \le 140\text{px}$, the track identity persists. An **occlusion grace period of 3.0 seconds** allows brief glances away or coffee sips without dropping the participant.

ATTENDANCE LEDGER COMMITS

The 10-Second Attendance Threshold

Transient hallway passersby or door monitors never trigger accidental attendance logs:

- **Duration < 3.0s:** Classified as `Transient`. Bounding bracket remains subtle roast/mocha.
- **Duration 3.0s – 9.9s:** Classified as `Verifying...`.
- **Duration ≥ 10.0s:** `Verified Present` badge unlocks. Bounding brackets glow warm caramel.
- **Automatic SQLite WAL Sync:** Writes row to `attendance_logs` table with first seen, last seen, and duration.

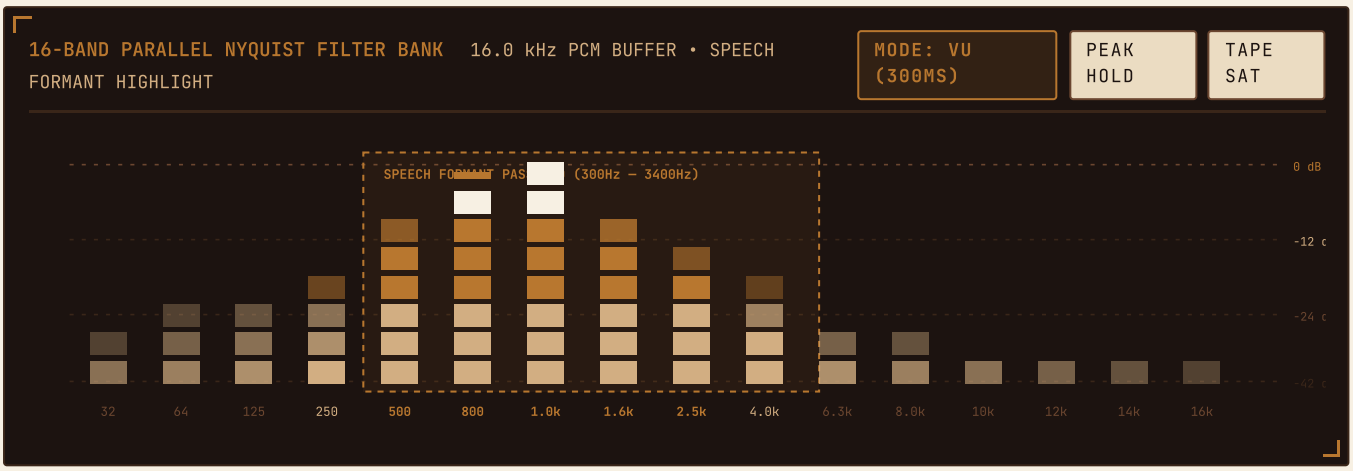
TAB [B] ATTENDANCE LEDGER // SCHEMA PREVIEW

PARTICIPANT ID	FIRST SEEN	LAST SEEN	SUSTAINED DURATION	STATUS	STORAGE ENGINE
Elena Rostova (Participant_01)	09:00:14 UTC	09:41:22 UTC	41m 08s	VERIFIED PRESENT	SQLite WAL Mode
Julian Sterling (Participant_02)	09:02:40 UTC	09:41:22 UTC	38m 42s	VERIFIED PRESENT	SQLite WAL Mode
Corridor Passerby (Participant_04)	09:14:02 UTC	09:14:05 UTC	2.8s	TRANSIENT (<3S)	Ephemeral Buffer

STEP-BY-STEP OPERATIONAL MILESTONE 03

Voice Frequency & The 16-Band Analog Spectrum Visualizer

Analyze vocal resonance via Librosa speech formant filters, VU ballistic physics, and vintage studio LED meter ladders.



VOICE ACTIVITY DETECTION (VAD)

Librosa Speech Formant Logic

The system isolates spoken classroom dialogue from ambient HVAC noise and shuffling through dual acoustic conditions:

```
is_speech = (300Hz ≤ Centroid ≤ 3400Hz) && (dBFS > -42.0 dB)
```

When active, the top badge illuminates: `VOICE ACTIVE (-23.4 dB)`. This signals the AI agent that interactive Socratic debate or student questions are occurring, preventing false inactivity penalties.

METER BALLISTICS MODES

Toggling Ballistic Behavior

Operators can click the meter mode pill in the upper right of the visualizer to adapt response physics:

- **VU Ballistic (Standard):** 300ms ANSI integration smoothing; models traditional needle inertia for comfortable visual pacing.
- **Peak Hold:** Instantaneous attack with a 75-frame latch; preserves peak energy crests with logarithmic gravity decay.
- **Tape Saturation:** Non-linear hyperbolic tangent ($\tanh$) soft compression curve.

PRACTICAL TESTING PROCEDURE

1. Normal Speech
Speak a complete sentence into your microphone. Watch the central bands (500Hz to 2.5kHz) light up into caramel and cream foam.

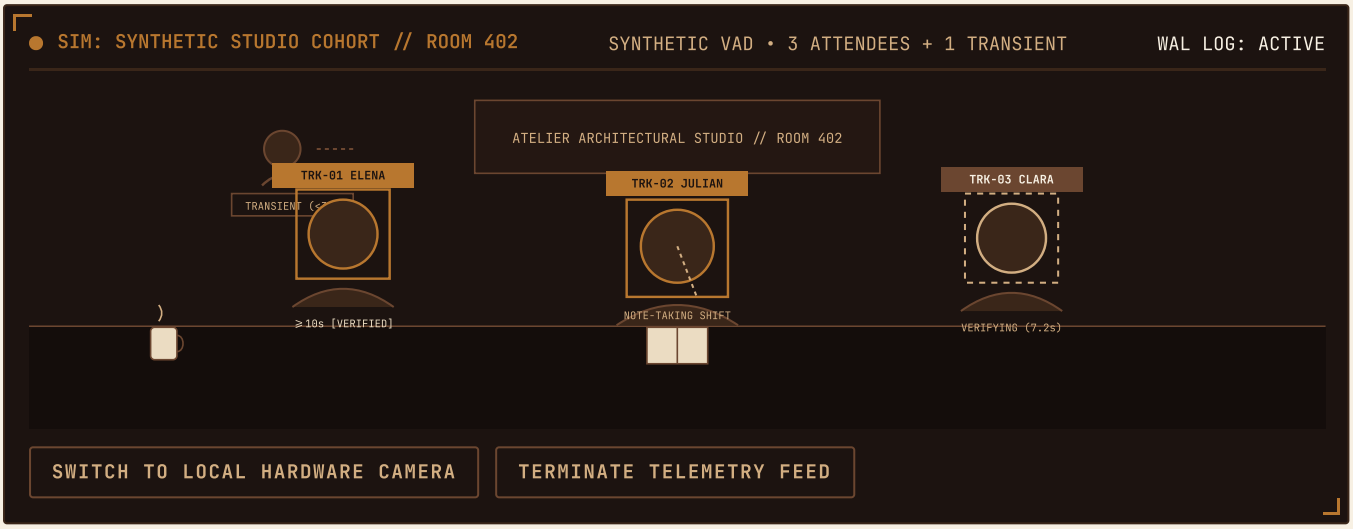
2. Ambient Noise Test
Remain silent while typing or tapping the desk. High-frequency transients remain below the speech centroid threshold; VAD remains idle.

3. Privacy Mute Test
Click [Mute Mic]. The entire 16-band ladder drops to idle floor (<0.05). Telemetry badge confirms `MIC MUTED`.

STEP-BY-STEP OPERATIONAL MILESTONE 04

Classroom Study Simulation (No Webcam Needed)

Deploy high-fidelity synthetic student avatars in Room 402 for privacy-restricted demos, lab evaluation, and algorithm verification.



SYNTHETIC AVATAR MATRIX

Room 402 Student Personas

- **Elena Rostova (TRK-01):** Seated at left desk. Continuously present from $t=0$; exhibits natural micro-sway (1.5Hz breathing) and frontal gaze focus (>90%).
- **Julian Sterling (TRK-02):** Seated at center desk. Every 16 seconds, enters a 4.5s note-taking shift with head dropping 14px toward his sketchbook to verify adaptive posture fusion.
- **Clara Vance (TRK-03):** Sits down at $t \ge 5.0\text{s}$. Clearly illustrates the live $0.0\text{s} \to 10.0\text{s}$ verification threshold in real time.

ANTI-FALSE-POSITIVE REJECTION

Transient Walk-by Verification

Every 26 seconds, a synthetic passerby walks across the corridor background:

- **Duration in Frame:** Exactly 4.2s ($t=3.0\text{s}$ to 7.2s).
- **Status:** Correctly rejected as `Transient (<3s)`.
- **Verification Guarantee:** Proves that students walking in and out of the classroom do not pollute the permanent SQLite ledger.

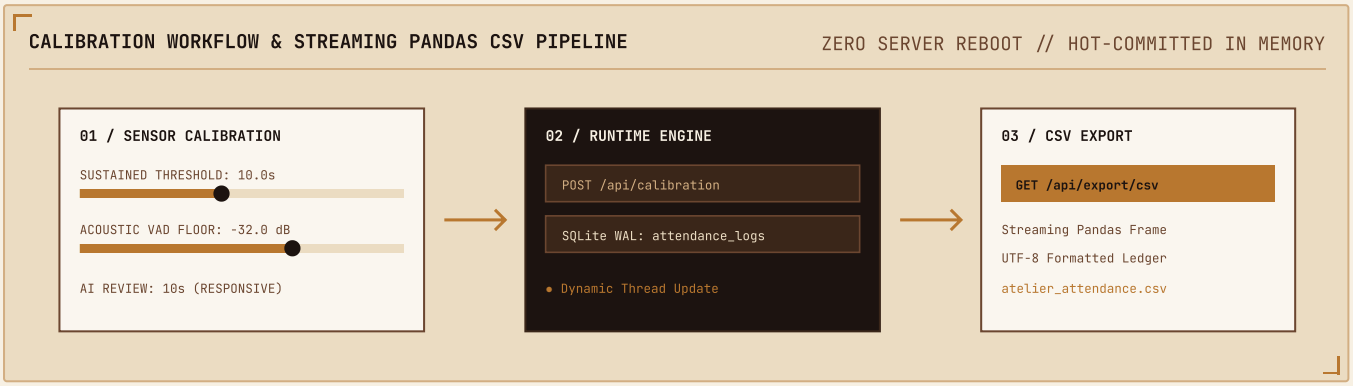
OPERATOR GUIDANCE // RUNNING BENCHMARKS WITHOUT HARDWARE

To run automated verification without a physical webcam, simply click `[Simulate Classroom Feed]`. The simulated feed outputs genuine OpenCV RGB frames and Librosa audio chunks, driving all dial gauges, sparklines, and AI pedagogical advice identically to live hardware.

STEP-BY-STEP OPERATIONAL MILESTONE 05

Engine Calibration & CSV Attendance Export

Tune temporal thresholds, adjust acoustic sensitivity floors, and stream verified SQLite records into pandas-formatted CSV files.



1 Tune Sustained Duration Slider (3s–30s)

Adjust how many uninterrupted seconds a student must remain tracked before achieving `Verified Present` status:

- **Fast Critique Sessions:** Set to `5.0s` for agile check-in.
- **Standard Studio Seminar:** Set to `10.0s` (Default).
- **High-Traffic Lecture Halls:** Set to `18.0s` to prevent hallway traffic false-positives.

2 Tune Acoustic VAD Floor (-50dB to -15dB)

Calibrate voice activation sensitivity to match your room acoustics:

- **Quiet Library / Seminar Room:** Lower to `-42.0 dB`.
- **Bustling Studio Workshop:** Raise to `-24.0 dB` to ignore tool noises.

3 Select AI Pedagogical Cadence

Choose the evaluation interval for pedagogical AI suggestions:

- **5 Seconds:** High-velocity feedback for fast studio pin-ups.
- **10 Seconds:** Balanced responsive rate (Recommended).
- **30 Seconds:** Contemplative pace for long-arc lectures.

4 Click [Commit Calibration] & [Export CSV]

Click `[Commit Calibration]` to immediately update the live Python engine. Then click `[Export Attendance Ledger (.CSV)]` to download a complete audit spreadsheet of all sessions.

DOWNLOADABLE CSV AUDIT COLUMNS // PANDAS SCHEMA

COLUMN	DATA TYPE	EXAMPLE ENTRY	AUDIT SIGNIFICANCE
session_id	STRING	Morning Architectural Seminar // Room 402	Identifies the course, studio section, and venue room.
participant_id	STRING	Participant_01	Persistent Euclidean tracking token assigned during stream.
first_seen	TIMESTAMP	09:00:14 UTC	Exact microsecond timestamp of initial face entry.
sustained_seconds	FLOAT	2468.2	Cumulative uninterrupted optical presence duration.
status	STRING	Verified Present (>=10.0s)	Accredited attendance verification outcome.

STEP-BY-STEP OPERATIONAL MILESTONE 06

Troubleshooting, Keyboard Shortcuts & FAQ

Quick-reference operator cards, keyboard shortcuts, resolution matrices for field anomalies, and privacy compliance assurance.

STUDIO KEYBOARD SHORTCUT CHEAT SHEET

Press '?' Toggle Concierge Guide	**Press 'Esc'** Minimize Guide / Dismiss	**Press 'Tab'** Cycle Controls & Sliders	**Space / Enter** Activate Selected Trigger

OPERATOR TROUBLESHOOTING PLAYBOOK

OBSERVED SYMPTOM	ROOT DIAGNOSTIC CAUSE	RECOMMENDED RESOLUTION
Webcam stream black / permission denied	Browser blocked video access, or camera claimed by Teams/Zoom/OBS.	Click the camera lock icon in Chrome address bar → select *Always Allow*. Terminate other conferencing apps, then re-click `[Initialize Local Camera]`.
Student tracked as multiple IDs	Sudden head movement exceeded distance threshold ($D > 140\text{px}$) or severe backlight.	Position lighting in front of attendees. Increase Euclidean gate in `face_detection.py` if working with wide-angle studio lenses.
Microphone visualizer shows no activity	Mic muted via UI toggle, system audio input disabled, or floor too high.	Check if `[Unmute Mic]` is shown. Verify system input in OS settings. Lower VAD sensitivity floor to `-42.0 dB` in Section 05.
Attendance not appearing in ledger	Participant presence duration did not reach sustained threshold.	Maintain continuous presence for ≥10 seconds until drafting bracket turns caramel and confirms `Verified Present`.

PRIVACY & COMPLIANCE

Frequently Asked Questions

Q: Are facial images or biometrics stored?

No. Optical frames are processed exclusively as ephemeral NumPy pixel arrays in volatile RAM. No faces, facial vectors, or video files are ever saved to disk.

Q: Can Atelier run without internet access?

Yes. All Haar cascades, Librosa FFT filters, and SQLite transactions execute locally. The pedagogical AI agent falls back to a deterministic 6-state offline heuristic engine if no API keys are present.

OFFICIAL CERTIFICATION SEAL

Verified Studio Rig

This operator manual certifies that the Atelier // Presence rig conforms to the Slow Morning editorial telemetry standard, offering complete pedagogical balance with strict student privacy.

ATELIER PRESENCE DIVISION
RELEASE: 2.1.0-STABLE
SQLITE WAL: SYNCHRONIZED

